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Questions

1. Notation for Families of Sets and Their Operations

Explain the notation used for the union, intersection, and Cartesian product of a family of sets Ai from i=1 to n.

$$\bigcup_{i=1}^{n} A_i$$

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Answers

1. Notation for Families of Sets and Their Operations

The notation denotes operations on a family of sets A_i for $i = 1$ to n:

$$\bigcup_{i=1}^{n} A_i = A_1 \cup A_2 \cup \cdots \cup A_n$$

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is the union of all A_i,

$$\bigcap_{i=1}^{n} A_i = A_1 \cap A_2 \cap \cdots \cap A_n$$

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is the intersection of all A_i, and

$$\prod_{i=1}^{n} A_i = A_1 \times A_2 \times \cdots \times A_n$$

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is the Cartesian product of all A_i. This notation summarizes the union, intersection, and Cartesian product of multiple sets indexed from 1 to n.
